

# Chapter 01 Homework — Groups, fields and vector spaces

Mikey Ferguson

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**Problem 01.3.** *Suppose that  $H$  is a subgroup of  $G$  of index 2. Show that  $H$  is a normal subgroup of  $G$ .*

We must show that  $ghg^{-1} \in H$  for every  $g \in G$  and every  $h \in H$ . Let  $h \in H$ .

**Case 1:**  $g \in H$ . Then  $ghg^{-1} \in H$  by closure; no normality is needed to guarantee this.

**Case 2:**  $g \notin H$ . Since  $H$  has index 2, there are exactly two cosets of  $H$  in  $G$ , namely  $H$  and one other; let  $gH$  be that disjoint coset of  $H$ , so  $g \notin H$ . Denote  $x = ghg^{-1}$ .

Suppose  $x \in gH$ . Then there exists  $h_1 \in H$  such that

$$ghg^{-1} = gh_1 \iff g^{-1} = h^{-1}h_1 \iff g \in H,$$

a contradiction.

So since  $x \notin gH$  due to the contradiction,  $x \in H$ , since  $H$  and  $gH$  partition  $G$ . So  $x = ghg^{-1} \in H$  and  $H$  is normal.  $\square$